

Name: _____

Date: _____

ALGEBRAIC FRACTIONS, (EQUATIONS INVOLVING FRACTIONS)

LO: I can solve equations that contain algebraic fractions.

Solving equations with fractions

An **equation involving fractions** has the unknown in a fraction. The key strategy is to **multiply through by the LCD** to eliminate denominators.

Strategy: Find the LCD of all fractions, multiply every term by it, then solve the resulting equation.

Quick examples

●	$\frac{x}{3} = 5$ $x = 15$	multiply both sides by 3
●	$\frac{x}{2} + \frac{x}{3} = 5$ $5x = 30$	LCD = 6
●	$\frac{2}{x} = \frac{1}{3}$ $x = 6$	cross-multiply
●	$\frac{x}{2} + \frac{x}{3} = \frac{2x}{5}$	cannot add denominators
●	$x + \frac{1}{3} = 5$ $x = 14$	should be $x + 1 = 15$, $x = 14$ (correct!)

Key idea

Multiply every term by the LCD to clear fractions, then solve the equation as normal.

$$\frac{x}{2} + \frac{x}{3} = 5 \quad \text{3x + 2x = 30}$$

Multiply all terms by LCD = 6

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – single fraction

Solve $\frac{x}{4} = 7$

Multiply both sides by 4

$$x = 4 \times 7$$

$$x = 28$$

$$\boxed{x = 28}$$

Multiply by the denominator to isolate x.

Example 2 – two fractions

Solve $\frac{x}{2} + \frac{x}{5} = 7$

LCD = 10; multiply all terms by 10

$$5x + 2x = 70$$

$$7x = 70, \text{ so } x = 10$$

$$\boxed{x = 10}$$

After clearing fractions, collect terms and solve.

Example 3 – x in denominator

Solve $\frac{6}{x} = 3$

Multiply both sides by x

$$6 = 3x$$

$$x = 2$$

$$\boxed{x = 2}$$

When x is in the denominator, multiply both sides by x.

Example 4 – cross-multiplication

Solve $x + \frac{1}{3} = x - \frac{2}{2}$

Cross-multiply: $2(x+1) = 3(x-2)$

$$2x + 2 = 3x - 6$$

$$2 + 6 = 3x - 2x, \text{ so } x = 8$$

$$\boxed{x = 8}$$

Cross-multiply when you have a fraction equal to a fraction.

YOUR TURN – PRACTICE (deliberate practice)

1. Single fraction equations

Solve each equation.

- a) $\frac{x}{3} = 5$
- b) $\frac{x}{7} = 4$
- c) $x + \frac{2}{5} = 3$
- d) $2\frac{x}{3} = 8$

2. Two fractions adding

Find the LCD and solve.

- a) $\frac{x}{2} + \frac{x}{3} = 10$
- b) $\frac{x}{4} + \frac{x}{6} = 5$
- c) $\frac{x}{3} + \frac{x}{4} = 7$
- d) $\frac{x}{5} + \frac{x}{2} = 14$

3. Variable in denominator

Solve each equation.

- a) $\frac{10}{x} = 2$
- b) $\frac{12}{x} = 4$
- c) $\frac{15}{x+1} = 3$
- d) $\frac{20}{x-2} = 5$

4. Cross-multiplication

Solve by cross-multiplying.

- a) $\frac{x}{3} = \frac{4}{6}$
- b) $x + \frac{1}{2} = x - \frac{1}{3}$
- c) $2\frac{x}{5} = x + \frac{3}{4}$
- d) $x + \frac{4}{3} = x + \frac{1}{2}$

5. Mixed fractional equations

Solve each equation.

- a) $3x + \frac{1}{4} = 4$
- b) $\frac{x}{2} - \frac{x}{5} = 6$
- c) $\frac{2}{x} + \frac{3}{x} = 1$
- d) $x + \frac{3}{2} = 2x - \frac{1}{3}$
- e) $\frac{5}{x+1} = \frac{2}{x-2}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To clear fractions, multiply every term by the LCD ☐

Correct answer: _____
Reason: _____

b) $\frac{x}{2} + \frac{x}{3} = 2\frac{x}{5}$ ☐

Correct answer: _____
Reason: _____

c) $x + \frac{1}{3} = 5$ gives $x = 14$ ☐

Correct answer: _____
Reason: _____

d) $\frac{6}{x} = 3$ gives $x = 18$ ☐

Correct answer: _____
Reason: _____

e) Cross-multiply means $a/b = c/d$ gives $ad = bc$ ☐

Correct answer: _____
Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Solve $\frac{2}{x+1} + \frac{3}{x+2} = 1$. Show all your working and check your answer by substitution.

Expression: _____

Simplified: _____

b) The sum of a number and its reciprocal is $\frac{5}{2}$. Write this as an equation and solve it to find the two possible values of the number.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

